2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2 Hours]

SCC.202 Human-Computer Interaction

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 20 marks.

Use of calculators is permitted.

Use a separate answer book for each question.

Question 1

1a. Figure 1 (below) shows a part of a user's desktop. The user has just finished renaming the file in the 'Source' folder. Hence it is still selected and the user's hands are homed on the keyboard. Now the user wants to copy the file to the 'Destination' folder on the left. There are two ways to do this.

Method 1: Drag and drop the file into the 'Destination' folder's space while holding down 'Ctrl' key.
Method 2: Use keyboard shortcuts only (e.g. Ctrl+C for copy).

Using predictive models (and the additional information provided below) identify which method will be faster. *(Show all computational steps)* **[10 marks]**

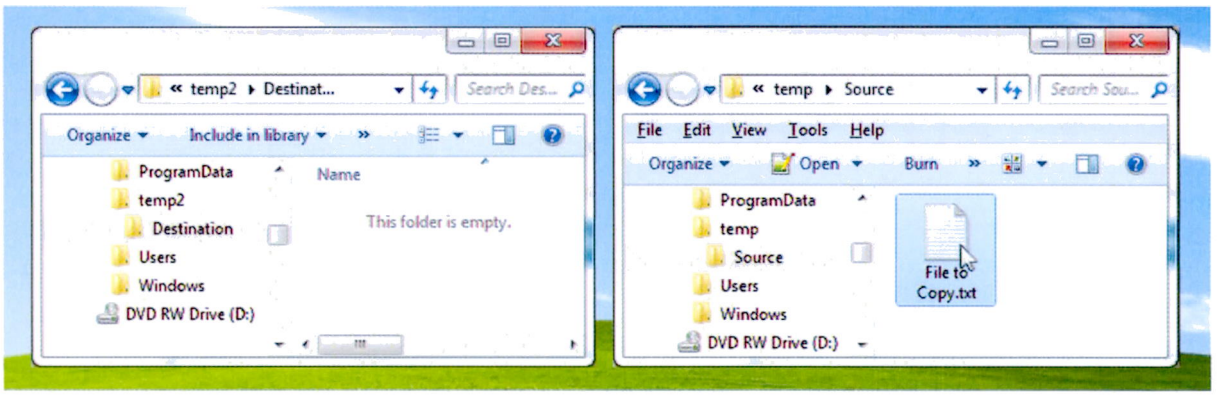

Figure 1: For Question 1a.

The following additional information is available:

i. Distance from current cursor position to edge of 'Destination' folder D = 400px
ii. Dimensions of 'Destination' folder space (width W, height H) W = 200px H = 100px
iii. Fitt's Law constants A = 0.3s; B = 0.2s
iv. Press or release mouse button (B_M) B_M = 0.1s;
v. Home hands to keyboard or mouse (H) H = 0.4s
vi. Mental act of thinking (M) M = 1.2s
vii. Keypress (K_P) K_P = 0.3s
viii. An Alt+TAB keypress will move the focus from one folder to another.

1b. You have been approached by a firm of Management Consultants to evaluate the usability of their new website and compare it with other, equivalent, websites. Write them an email in which you outline the different approaches to usability evaluation – in particular explain the difference between an **empirical** user study and an analytic or **heuristic** evaluation. In each case explain how you would go about conducting a usability study and the advantages and disadvantages of each approach. **[10 marks]**

Total 20 marks

Question 2

2a. What are TUIs? Based on Ullmer's MetaDesk (1997), sketch three TUI metaphors for standard graphical UI elements. **[5 marks]**

2b. Describe the ways in which a pie chart encodes values being plotted. Also discuss why pie-charts should not be used for displaying data containing more than two variables. **[5 marks]**

2c. An airline company has written to you suggesting that they are especially concerned about the possibilities for 'dark patterns' in their website and that this might represent instances of 'bad' usability. Write an email explaining what a 'dark pattern' is, the kinds of 'dark patterns' that can exist on websites, and how this might impact on usability. **[10 marks]**

Total 20 marks

Question 3

3a. Using one of the following TUI examples, explain how tangibility can enhance the execution of an interactive task? **[5 marks]**
TUI Examples: ZeroN, GaussBricks, Siftables, Reactable.

3b. Analyse the game of Tetris from the perspective of epistemic actions as well as pragmatic actions. **[5 marks]**

3c. You are part of a large European research consortium submitting a proposal for HCI research to the European Commission. You are specifically tasked with covering the 'ethical issues' component of the proposal and to prepare a short document for your consortium partners explaining some of the most important 'ethical issues' that need to be considered. Explain what is meant by the notion of '**informed consent'**, what specific promises you need to make to your research subjects/participants that might form part of the idea of 'informed consent', and what problems might arise before, during and after the research. **[10 marks]**

Total 20 marks

Question 4

4a. Explain using examples (of experimental design) as to when you would use a directional (one-tailed) t-test and a non-directional (two-tailed) t-test? **[4 marks]**

4b. Describe the relationship of Type-I and Type-II errors with the directionality of the test (one-tailed or two-tailed) at same significance levels. A selection of values is provided from the t-stat table to help formulate the answer.

[6 marks]

(Df=10) *α-values* →	*0.05*	*0.02*	*0.01*
One-tailed t_{crit}	*1.812*	*2.228*	*2.764*
Two-tailed t_{crit}	*2.228*	*2.764*	*3.169*

4c. You are part of a large European research consortium submitting a proposal for HCI research to the European Commission. You are specifically tasked with covering the 'ethical issues' component of the proposal and to prepare a short document for your consortium partners explaining some of the most important 'ethical issues' that need to be considered. Specifically:

i. Some of your consortium partners work in academia and others in industry – identify and comment on some of the **ethical codes of conduct** that govern their everyday work.

[5 marks]

ii. Using the example of the Facebook 'emotional contagion' case – outline to your European partners what the difference might be between 'normal business practice' and 'unethical research practice' **[5 marks]**

Total 20 Marks

---END OF PAPER---